CA Theory PSQA

1)Distinguish between Computer Architecture and Computer Organization. (4M)

Answer:

Parameter	Computer Architecture	Computer Organization
Definition	Attributes visible to the programmer (logical aspect)	Operational units & interconnections (physical realization)
Focus	What the computer does (ISA, instruction set)	How the hardware executes tasks (ALU design, buses)
Examples	Instruction formats, Addressing modes, Registers	Control signals, Memory technology, Clock speed
Change Rate	Changes rarely (e.g., x86, ARM64)	Upgraded frequently across processor generations
Visibility	Directly visible to assembly programmer	Transparent/hidden from assembly programmer

2)**Briefly discuss the evolution of computers and how these changes impacted computer performance and usability.**

Answer:**Brief Evolution of Computers & Their Impact:** Computers evolved through five main generations, each making machines faster, smaller, cheaper, and easier to use.

Evolution of Computers

- 1st Generation – Vacuum Tubes Big, slow, heated up quickly. Used for basic scientific work. Impact: Very limited performance and hard to use.
- 2nd Generation – Transistors Smaller, faster, more reliable than vacuum tubes. Impact: Improved speed and usability; computers became practical for business.
- 3rd Generation – Integrated Circuits Many transistors on one chip → huge jump in speed and efficiency. Impact: Computers became more powerful and affordable.
- 4th Generation – Microprocessors Entire CPU on a single chip → personal computers (PCs) emerged. Impact: Computers entered homes; usability increased massively.
- 5th Generation – AI & Parallel Processing Modern systems use AI chips and parallel processing. Impact: Extremely high performance, smart features, easy interfaces.

Overall Impact on Performance & Usability

- **Speed increased** from slow vacuum tubes to ultra-fast AI processors.
- **Size decreased** from room-sized machines to handheld devices.
- **Reliability improved** with transistors and ICs.
- **Cost dropped**, making computers widely accessible.
- **Usability improved** with PCs, mobile devices, and user-friendly interfaces.
- **Modern computers** support AI, automation, and complex tasks effortlessly.

One-line summary- Computers evolved from large, slow machines to fast, smart, user-friendly devices — improving performance and making technology accessible to everyone.

3)**Explain the basic architecture of a computer system including its main components.**

Answer: **Basic Architecture of a Computer System**: A computer system has five main components that work together to process data.

Main Components of Computer Architecture

Input Unit	Takes data and instructions from the user (keyboard, mouse, scanner).
Output Unit	Shows results to the user (monitor, printer).
Memory Unit	Stores data and instructions. Includes Primary Memory (RAM, ROM) and Secondary Storage.
ALU	Performs all arithmetic and logical operations
Control Unit	Directs and coordinates all parts of the computer.

Easy Summary

- Input → Process (ALU + CU) → Output
- Memory stores everything needed for processing.

One-line revision: A computer system works by taking input, storing data, processing it using ALU & CU, and giving output.

3)**Define a multiprocessor. Explain the characteristics of multiprocessors connected by a network.**(4M)

Answer: **Multiprocessor:** A **multiprocessor** is a computer system that has **two or more processors (CPUs)** working together. These processors share tasks to increase speed, reliability, and overall performance.

Characteristics of Multiprocessors Connected by a Network

- **Independent Processors** Each processor works on its own tasks but can communicate with others through a network.
- **Separate Memory** Every processor usually has **its own local memory**, reducing conflicts and improving speed.
- **Message Passing** Processors exchange data using **network messages** instead of shared memory.
- **High Scalability** More processors can be added easily because they are connected through a network.
- **Fault Tolerance** If one processor fails, others continue working, making the system more reliable.

Easy Summary: Multiprocessor = Many CPUs working together. Network-connected multiprocessors = Independent CPUs + own memory + communicate via messages + scalable + reliable.

5)**What is virtual memory? Highlight different types of virtual memory .(4M)**
Answer: **Virtual Memory**: Virtual memory is a technique that allows a computer to use more memory than the physical RAM available. It does this by using a part of the hard disk as extra memory, so programs can run smoothly even if RAM is full.

Definition : Virtual memory = RAM + part of hard disk used as temporary memory It gives the illusion of a large memory to the CPU.

Types of Virtual Memory

- **Paging** Memory is divided into fixed-size blocks called *pages*. If RAM is full, pages are moved between RAM and disk.
- **Segmentation** Memory is divided into variable-size segments (code, data, stack). Helps manage programs more logically.
- **Demand Paging** Only required pages are loaded into RAM when needed. Saves memory and reduces loading time.

Easy Summary: Virtual memory = Uses hard disk as extra RAM Types = Paging, Segmentation, Demand Paging

6)**Demonstrate the working procedure of direct memory access (DMA) on a computer system. What are the advantages of it ? (6M)**
Answer: DMA Working Procedure : Direct Memory Access (DMA) allows data to move between memory and I/O devices without using the CPU for every step. This makes data transfer faster and more efficient.

Working Procedure of DMA

1. CPU initializes DMA CPU gives DMA the source address, destination address, and word count.
2. DMA requests the bus DMA sends a bus request signal to take control of the system bus.
3. CPU grants the bus CPU stops using the bus and gives control to DMA (bus grant).
4. DMA transfers data directly DMA moves data from device → memory or memory → device without CPU involvement.
5. DMA releases the bus After finishing, DMA returns control to the CPU.
6. DMA interrupts CPU DMA sends an interrupt to tell the CPU that the transfer is complete.

Advantages of DMA

- Faster data transfer — moves large blocks of data quickly.
- Reduces CPU workload — CPU is free to do other tasks.
- Improves overall system performance — avoids bottlenecks in data movement.
- Efficient for high-speed devices — like disks, sensors, network cards.
- Better multitasking — CPU continues running programs while DMA works.

Easy Summary: DMA = CPU sets up → DMA transfers data → CPU continues working → DMA signals done.

8)**Define interrupt-driven I/O. Analyze how it uses an interrupt signal to improve CPU efficiency**

Answer: Interrupt-Driven I/O: Interrupt-driven I/O is a method where I/O devices send an interrupt signal to the CPU when they are ready for data transfer. This avoids the CPU constantly checking devices and improves efficiency.

Definition: Interrupt-driven I/O is a technique where the CPU waits for an interrupt from an I/O device instead of continuously monitoring it. When the device is ready, it interrupts the CPU to start data transfer.

Working Procedure

1. CPU starts an I/O operation and continues doing other tasks.
2. The I/O device works independently in the background.
3. When the device is ready (data available or task completed), it sends an interrupt signal to the CPU.
4. The CPU pauses its current work, saves its state, and handles the I/O request.
5. After completing the I/O service, the CPU returns to its previous task.

How Interrupts Improve CPU Efficiency

- No continuous checking (polling) → CPU is free for other work.
- CPU only responds when needed → reduces wasted time.
- Faster overall performance because CPU and I/O work in parallel.
- Better multitasking → CPU handles multiple processes smoothly.
- Efficient for slow devices → CPU doesn't wait idly.

Easy Summary: Interrupt-driven I/O = Device alerts CPU → CPU handles it → CPU continues work. This saves time and makes the system faster and more efficient.

9)**Explain the basic architecture of a computer system including its main components.**

Answer: **Basic Architecture of a Computer System** :A computer system is built from five main components that work together to take input, process it, store it, and produce output.

Main Components of Computer Architecture

Input Unit	Devices like keyboard and mouse that send data to the computer.
Output Unit	Devices like monitor and printer that display results.
Memory Unit	Stores data and instructions. Includes Primary Memory (RAM, ROM) and Secondary Storage.
CPU	The brain of the computer. It has: ALU (calculations) and CU (control and coordination).

System Bus	A communication path that connects all components (data bus, address bus, control bus).

How the System Works

1. Input Unit gives data to the computer.
2. Memory Unit stores the data and instructions.
3. CPU processes the data using ALU and CU.
4. Output Unit shows the final result.
5. System Bus carries signals between all parts.

Quick Summary: Input → Memory → CPU → Output All connected through the system bus.

11) **How does the cache memory work in a computer system? Briefly describe Pentium-4 Cache organization . (6M)**

Answer: Cache Memory Working & Pentium-4 Cache Organization

1. How Cache Memory Works

Cache memory is a small, very fast memory placed between the CPU and main memory (RAM). It stores the data and instructions that the CPU is most likely to need soon.

Working procedure :

1. CPU needs some data → First checks the Cache.
2. Cache Hit: If the data is found in cache → CPU takes it immediately (very fast).
3. Cache Miss: If data is not in cache → Data is brought from main memory (RAM) into cache, and then given to CPU.
4. Cache uses the principle of Locality of Reference (programs usually use the same data/instructions again and again).

Simple way to remember: Cache = Super-fast temporary notebook of the CPU.CPU first looks in the notebook (cache). If found → Hit. If not → Go to the big book (RAM).

2. Pentium-4 Cache Organization (Brief) Pentium-4 uses a multi-level cache system:

Level	Name	Size (typical)	Special Feature
L1	Level-1 Cache	8 KB (Data) + 12K μops (Instruction Trace Cache)	Very fast, closest to CPU cores
L2	Level-2 Cache	256 KB or 512 KB	Unified (stores both data & instructions)
L3	Level-3 Cache	Present in some models (later versions)	Shared between cores

Key points of Pentium-4 Cache:

- Uses Trace Cache for instructions (stores decoded instructions instead of normal instructions → faster execution).

- L1 Data Cache is 8-way set associative.
- L2 Cache is 8-way set associative and uses Advanced Transfer Cache technology for high bandwidth.
- Cache is organized to support the high clock speeds of Pentium-4.
- Quick Summary (Easy to remember for 6 marks)
- Working: CPU → Cache (Hit/Miss) → RAM
- Pentium-4: Multi-level (L1 Trace Cache + L1 Data + L2 Unified) with high associativity and Trace Cache technology.

13)**What is memory hierarchy in Computer Architecture ? Explain the significance of it . (5M)**

Answer: Memory Hierarchy in Computer Architecture : Memory hierarchy is the arrangement of different types of memory in a computer system based on speed, cost, and size. It helps the CPU access data faster and more efficiently.

Definition :Memory hierarchy is a structure where memory is organized from fastest & smallest to slowest & largest, so the CPU gets data quickly while keeping storage affordable.

Levels of Memory Hierarchy

- Registers — Fastest, inside CPU.
- Cache Memory — Very fast, stores frequently used data.
- Main Memory (RAM) — Medium speed, stores active programs.
- Secondary Storage — Slow but large (HDD, SSD).
- Tertiary Storage — Very slow, used for backups.
- Significance of Memory Hierarchy (Simple & clear)
- Improves speed — CPU gets data from faster memory first.
- Reduces cost — Combines expensive fast memory with cheaper large memory.
- Efficient data access — Frequently used data stays in cache/registers.
- Better performance — Less waiting time for CPU → faster execution.
- Optimizes storage — Balances speed, size, and cost effectively.

Easy Summary: Memory hierarchy = Fast small memory at top + slow large memory at bottom → better speed and lower cost.

14)**Define programming I/O. Discuss different types commands in programming I/O. (5M)**

Answer: Programming I/O: Programming I/O means controlling input/output operations using software instructions. The CPU uses specific I/O commands to communicate with devices like keyboard, printer, disk, etc.

Definition: Programming I/O is a method where the CPU uses I/O instructions to transfer data between memory and I/O devices. All control is done by the program, not by hardware automatically.

Types of Commands in Programming I/O

- Input Command — Transfers data from I/O device to CPU. Example: CPU reads a character from keyboard.
- Output Command — Sends data from CPU to I/O device. Example: CPU sends data to printer.
- Status Command — Checks the status of an I/O device (busy, ready, error). Helps CPU know when the device can accept or send data.
- Control Command — Sends control signals to I/O devices. Example: Start, stop, reset, or initialize a device.

Easy Summary: Programming I/O = CPU controls I/O using commands → Input, Output, Status, Control.

15) **What is BUS? Shortly explain the types of buses used in Computer Architecture. (3M)**
Answer: BUS in Computer Architecture: A bus is a communication pathway that carries data, addresses, and control signals between different parts of a computer system.

Types of Buses

- Data Bus — Carries actual data between CPU, memory, and I/O devices.
- Address Bus — Carries memory addresses to locate where data should be read or written.
- Control Bus — Carries control signals like read, write, interrupt, and timing signals.

Easy Summary: Bus = Pathway for communication Types = Data (what), Address (where), Control (how)

16)**Difference between ALU and Logic operation with example. (3M)**
Answer: Difference Between ALU and Logic Operation (Simple 3-mark answer)

ALU	Logic Operation
ALU = Arithmetic Logic Unit inside the CPU.	Logic operations are specific functions performed *by* the ALU.
Performs arithmetic + logic tasks.	Performs true/false decisions.
Example: 5 + 3, 10 − 2, AND, OR.	Example: A AND B, A OR B, NOT A.

Easy Summary:

- ALU = Hardware unit that performs arithmetic + logic.
- Logic operations = Only the logical part (AND, OR, NOT).
- Example to remember
- ALU example: 7 + 2
- Logic operation example: A AND B

17)**Define CPU and its major parts (3M)**

Answer: CPU and Its Major Parts: The CPU (Central Processing Unit) is the brain of the computer. It performs all processing tasks and controls the entire system.

Major Parts of CPU

ALU	Performs arithmetic (add, subtract) and logic (AND, OR) operations.
Control Unit	Controls and coordinates all activities inside the computer.
Registers	Small, fast memory inside CPU used to store temporary data.

Easy Summary: CPU = ALU (calculations) + CU (control) + Registers (fast storage)

18)**Explain the role of buses in a computer system including the functions of the data bus, address bus and control bus. How do these components contribute to the interaction between the processor and other hardware? (5M)**

Answer: Role of Buses in a Computer System: A bus is a communication pathway that connects the processor, memory, and I/O devices. It carries data, addresses, and control signals so all parts of the computer can work together smoothly.

Types of Buses and Their Functions

- Data Bus Carries actual data between CPU, memory, and I/O devices. *Example:* Sending a number from RAM to CPU.
- Address Bus Carries memory addresses that tell where data should be read or written. *Example:* CPU selects a specific memory location.
- Control Bus Carries control signals like read, write, interrupt, and timing signals. *Example:* CPU tells memory whether it wants to read or write.
- How These Buses Help Processor–Hardware Interaction
- The CPU uses the address bus to choose the exact memory location.
- The CPU uses the data bus to send or receive data.
- The CPU uses the control bus to manage operations (read/write/interrupt).
- Together, they allow the CPU to communicate efficiently with memory and I/O devices.
- This ensures smooth data flow, faster processing, and proper coordination inside the computer.

Easy Summary: Data Bus = What Address Bus = Where Control Bus = How All three buses work together to let the CPU interact with memory and devices correctly and efficiently.

19)**Explain with an example why modern systems often prefer serial communication over parallel despite its slower theoretical data rate. (5M)**

Answer: Why Modern Systems Prefer Serial Communication: Modern computers often choose serial communication even though parallel communication *seems* faster. This is because serial communication is more reliable, cheaper, and works better over long distances.

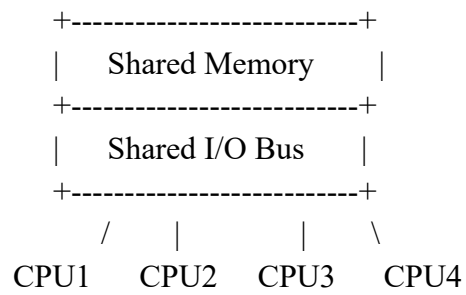
Reason: Parallel communication sends many bits at once through multiple wires. Serial communication sends one bit at a time through a single wire.
Even though parallel looks faster, it has major problems in modern high-speed systems.

Why Serial is Preferred

- No Skew Problem In parallel communication, signals on different wires don't arrive at the same time (called *skew*). Serial avoids this because it uses only one line, so timing stays perfect.
- Long-distance reliability Parallel signals get distorted over long cables. Serial works better and cleaner over long distances.
- Cheaper and simpler wiring Parallel needs many wires → expensive and bulky. Serial needs just one pair → cheaper and easier.
- Higher real-world speed Modern serial technologies (USB, SATA, PCIe) use high-frequency signaling, making them faster than old parallel systems.
- Less interference Fewer wires = fewer chances of electrical noise.

Example: Old parallel ATA (PATA) used 40–80 wires and had speed problems due to skew. It was replaced by Serial ATA (SATA), which uses just 7 wires and is much faster and more reliable.
Easy Summary: Serial communication = fewer wires, no timing issues, cheaper, cleaner signals, and faster in practice.

20)**Consider a system with four processors sharing memory and I/O resources. Design a simple block diagram showing symmetric multiprocessing (SMP) and explain how tasks are handled and resources are shared. Then, describe how the same would differ in a distributed multiprocessing system.** (6M)
Answer: SMP vs Distributed Multiprocessing
Symmetric Multiprocessing (SMP): In SMP, all processors share the same memory and I/O. They work together as equals.
Simple Block Diagram



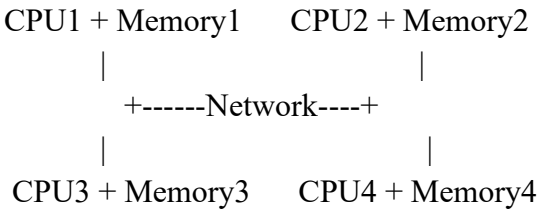
How SMP Works

- All four processors share the same memory and I/O devices.
- The operating system assigns tasks to any processor.
- If one CPU is busy, another CPU can take the task.
- Processors communicate through shared memory.
- Resources like RAM and I/O are common, so coordination is needed.

- Task Handling in SMP
- Tasks are placed in a common task queue.
- Any processor can pick up any task.
- Shared memory allows fast communication and easy data sharing.
- Good for servers, multi-core CPUs, and parallel processing.

Distributed Multiprocessing (How it differs): In distributed multiprocessing, each processor has its own memory and sometimes its own I/O.

Simple Block Diagram



Key Differences

- Each processor has separate memory → no sharing.
- Communication happens through a network (messages).
- Tasks are divided among processors, but data sharing is slower.
- More scalable: you can add more processors easily.
- Used in clusters, cloud systems, and distributed servers.

Easy Summary

- SMP: Shared memory, shared I/O, processors work as equals.
- Distributed: Each processor has its own memory; communication via network.
- SMP = Fast sharing, Distributed = High scalability.

21)**Describe how cache memory works in a computer system. Explain what happens when the CPU requests data include the concepts of cache hit , cache miss and how data is fetched from memory .Use a simple example to support your explanation.** (6M)

Answer: Cache Memory Working: Cache memory is a small, very fast memory that stores data the CPU uses often. It helps the CPU work faster by reducing the time needed to access data from main memory (RAM).

How Cache Memory Works

1. CPU requests data

When the CPU needs data, it first checks the cache instead of RAM.

2. Cache Hit

If the required data is found in the cache, it is called a cache hit.

- CPU gets the data immediately → very fast.

3. Cache Miss

If the data is not found in the cache, it is called a cache miss.

- CPU then fetches the data from main memory (RAM).

- The fetched data is also stored in the cache for future use.

4. Fetching Data from Memory

When a miss happens:

- CPU sends an address through the address bus.
- RAM sends the required data through the data bus.
- Cache stores this data because it might be needed again soon.

Simple Example

Imagine the CPU needs the value stored at memory location 1000.

Case 1: Cache Hit

- CPU checks cache → location 1000 is already there.
- CPU reads it instantly → fast performance.

Case 2: Cache Miss

- CPU checks cache → location 1000 is not there.
- CPU fetches data from RAM.
- Cache stores this data.
- Next time CPU asks for location 1000 → cache hit → faster.

Why Cache Makes the System Faster

- Cache is much faster than RAM.
- Most programs reuse data → cache stores it.
- CPU spends less time waiting, improving overall performance.

Easy Summary: Cache Hit = Data found → fast Cache Miss = Data not found → fetch from RAM → slow Cache stores frequently used data to make the CPU work faster..